

Statistics II

CSIT — 3rd Semester — 2080 — Answer Sheet
Subject Code: STA215

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data and how many tables are used to arrange each data set. Efficiency will be measured the number of processed requests per hour. Applying the program to data set of different sizes and number of tables are used, she gets the following results. Processed requests, Y 16261141505540 Data size, (gigabytes), X_1 1510108776 Number of tables, X_2 12101020204 The regression equation obtained is: $Y = 52.7 - 2.87 X_1 + 0.85 X_2$. Total sum of square = 1452 Sum of due to regression = 1143.3. a) Interpret the values of regression coefficients b_1 and b_2 b) Test the significance of the regression model at 0.05 level of significance. c) Is there significant relationship between processed request and number of tables at 0.05 level of significance? Given, standard error of $b_1 = 0.55$. d) What percentage of variation of processed requests is explained by data size and number of tables? e) Compute standard error of estimate. f) Estimate the number of processed requests if data size is 9 gigabytes and number of tables used are 8.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. In an experiment to determine which of three different missile systems is preferable, the propellant burning rate is measured. The data after coding are given in the table. Use Kruskal-Wallis test (significance level of 0.01) to test the hypothesis that the propellant burning rates are same for the three missile systems. Missile system I 22.316.722.719.318.5 Missile system II 23.419.517.520.816.019.9 Missile system III 18.419.517.818.019.622.817.1

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is the Latin Square Design? Under what conditions can this design be used? Give the lay out and analysis of Latin Square Design.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. What do you understand by estimation? If we want to determine average mechanical aptitude of a large group of workers, how large a random sample is need to be able to assert with probability 0.95 that the sample mean will not differ from the true mean by more than 2.0 points? Assume that population standard deviation is 10.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define level of significance. Describe type I and type II errors.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. A random sample of students is asked their opinion on proposed core curriculum change. The results are as follows:

Class	Opinion Favoring	Opinion Opposing
Freshman	120	80
Sophomore	60	140
Junior	50	50
Senior	40	53

Test the hypothesis that opinion on the change is independent of class standing. Use 0.01 significance level.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define Central limit theorem. The life of a certain brand of an electric bulb may be considered a random variable with mean 1350 hours and standard deviation 550 hours. Using central limit theorem, find the probability that the average life time of 100 bulbs exceeds 1440 hours.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define multiple correlation. In a trivariate distribution X_1 , X_2 and X_3 , the simple correlation coefficients are given as $r_{12} = 0.5$, $r_{13} = 0.6$, $r_{23} = 0.7$. Find (i) partial correlation coefficient between X_1 and X_2 keeping X_3 constant. (ii) multiple correlation coefficient assuming X_3 as dependent variable.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What do you understand by Design of Experiment? Prepare one way analysis of variance table and carry out the test for the significance of difference in the average yields between three different varieties of seed. Given: Total sum of squares = 258, Sum of square between varieties of seed = 50, Total number of observations = 20

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define type I and type II error in testing of hypothesis. It is claimed that Samsung and Huawei mobiles are equally popular in Kathmandu. A random sample of 600 people from Kathmandu showed 350 have Samsung mobile. Test the claim at 5% level of significance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Customers of certain internet service provider connect to the internet at the average rate of 10 new connections per minute. Connections are modelled by binomial counting process.a) What frame length gives the probability 0.1 of an arrival during given frame?b) Find the mean and variance for the number of seconds between two consecutive connections.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on any two:a) Difference between parametric and non-parametric test.b) Required assumptions for linear regression model.c) Stochastic process.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.